

Logical Gates by Gauge-Field Formalism of Quantum Error Correction

Junichi Haruna (Kyoto U.), based on arXiv:2511.15224



0. Summary

- We construct **logical gates for general CSS codes** using the gauge-field formalism.
- Logical gates are written as **exponential of polynomials of electric and magnetic gauge fields**, which directly give physical-gate decompositions.
- Applies to general CSS codes without assuming manifold or cup-product structures.
- Covers **phase gates, their multi-controlled versions and Hadamard gate**
- Logical actions depend only on the corresponding (co)homology classes.

1. Motivation

- QEC protects logical qubits from noise.
- Universal fault-tolerant computation requires logical gates on encoded qubits.
- Existing constructions often rely on special code families or geometric structures.
- General CSS codes can be described by chain complexes.
- Gauge field formalism expresses logical operators as homology/cohomology-based gauge-field operators.

Key question

Can the gauge field formalism give a systematic construction of logical gates for general CSS codes?

2. CSS codes and gauge fields

- Chain-complex representation** of CSS codes:

$$C_2 \rightarrow C_1 \rightarrow C_0, \quad \partial_1 \partial_2 = 0$$

with boundary maps $\partial_2 = H_Z^T$, $\partial_1 = H_X$.

- Qubits on 1-chains; stabilizers on 2-/0-chains;

$$S^Z(f) = \bar{Z}(\partial_2 f), \quad S^X(v) = \bar{X}(\partial_1^T v).$$

- Logical Pauli operators as (co)homology groups:

$$H_1 = \ker \partial_1 / \text{Im } \partial_2, \quad H^1 = \ker \partial_2^T / \text{Im } \partial_1^T.$$

- Electric / magnetic gauge fields:**

$$a(e_j) = z_j, \quad b(e_j) = x_j, \quad Z_j = e^{i\pi z_j}, \quad X_j = e^{i\pi x_j}.$$

- Transversal Pauli operator = Wilson loop

$$\bar{Z}(\gamma) = e^{i\pi a(\gamma)} = \prod_j Z_j^{\gamma_j}, \quad \bar{X}(\gamma) = e^{i\pi b(\gamma)} = \prod_j X_j^{\gamma_j}$$

with $\gamma = \sum_i \gamma^i e_i$ ($\gamma^i = 0, 1$)

Key idea

Logical gates = functions of gauge fields $a(\gamma)$ and $b(\tilde{\gamma})$

3. Construction principle

- Any logical Pauli function $\rightarrow$ logical operator:

$$[g(\bar{X}, \bar{Z}), S^Z(f)] = [g(\bar{X}, \bar{Z}), S^X(v)] = 0.$$

- Lift physical gate by replacement:

$$\mathcal{O} = g(X, Z) \Rightarrow \bar{\mathcal{O}} = g(\bar{X}, \bar{Z}).$$

- Gauge fields $\rightarrow$ **physical-gate decompositions**

$$Z^c = I - 2cz, \quad X^c = I - 2cx, \quad c = 0, 1.$$

- Expand polynomial in $a(\gamma)$, $b(\tilde{\gamma}) \rightarrow$ explicit gates

$$\bar{Z}(\gamma) = e^{i\pi a(\gamma)}, \quad \bar{X}(\tilde{\gamma}) = e^{i\pi b(\tilde{\gamma})}$$

- Logical action depends only on **(co)homology classes**

$$\mathcal{O}(\gamma) \sim \mathcal{O}(\gamma + \partial f) \text{ on the code space}$$

- Purely algebraic; works for **general CSS codes** (no geometry / cup products / locality)
- No fault-tolerance / depth optimization here – general decomposition only

4. Main Result ($\gamma, \gamma_i \in H_1, \tilde{\gamma} \in H^1, \langle \gamma, \tilde{\gamma} \rangle = 1$)

Gate type	Physical decomposition	Gauge-field expression
$\bar{S}$	$\prod_k S_{i_k} \prod_{k_1 < k_2} CZ_{i_{k_1}, i_{k_2}}$	$e^{i\frac{\pi}{2}a(\gamma)^2}$
$\bar{H}$	$\bar{S}(\gamma) H(\tilde{\gamma}) \bar{S}(\tilde{\gamma}) H(\tilde{\gamma}) \bar{S}(\gamma)$	$e^{i\frac{\pi}{2}a(\gamma)^2} e^{i\frac{\pi}{2}b(\tilde{\gamma})^2} e^{i\frac{\pi}{2}a(\gamma)^2}$
$\bar{CZ}$	$\prod_{k_1, k_2} CZ_{i_{k_1}^1, i_{k_2}^2}$	$e^{i\pi a(\gamma_1)a(\gamma_2)}$
$\overline{C^{m-1}Z}$	$\prod_{k_1, \dots, k_m} C^{m-1}Z_{i_{k_1}^1, \dots, i_{k_m}^m}$	$e^{i\pi a(\gamma_1) \cdots a(\gamma_m)}$
$\bar{T}$	$\prod_k T_{i_k} \prod_{k_1 < k_2} CS_{i_{k_1}, i_{k_2}}^\dagger$ $\times \prod_{k_1 < k_2 < k_3} CCZ_{i_{k_1}, i_{k_2}, i_{k_3}}$	$e^{i\frac{\pi}{4}(2a(\gamma)^3 - 3a(\gamma)^2 + 2a(\gamma))}$

5. Example: T gate

- Physical T gate (**non-Clifford phase**):

$$T = \exp\left(i\frac{\pi}{4}\frac{I - Z}{2}\right)$$

- Replace $Z \rightarrow$ logical $\bar{Z}(\gamma) \rightarrow$ define

$$\bar{T}(\gamma) = \exp\left(i\frac{\pi}{4}\frac{I - \bar{Z}(\gamma)}{2}\right)$$

- Expand $\bar{Z}(\gamma) = e^{i\pi a(\gamma)} \rightarrow$ **cubic gauge-field polynomial**:

$$\bar{T}(\gamma) = \exp\left[i\frac{\pi}{4}(2a(\gamma)^3 - 3a(\gamma)^2 + 2a(\gamma))\right]$$

- Physical decomposition ($\text{supp}(\gamma) = \{i_1, \dots, i_m\}$):

$$\bar{T}(\gamma) = \prod_k T_{i_k} \prod_{k_1 < k_2} CS_{i_{k_1}, i_{k_2}}^\dagger \prod_{k_1 < k_2 < k_3} CCZ_{i_{k_1}, i_{k_2}, i_{k_3}}$$

- Well-definedness on the code space:

$$\bar{T}(\gamma) \sim \bar{T}(\gamma + \partial_2 f)$$

6. Conclusion and outlook

Main message

- Gauge fields = compact language **linking logical gates, CSS homology & lattice gauge theory**
- Systematic logical gates for general CSS codes
- Purely algebraic – no geometry / locality / cup products / special codes

Future directions

- Add fault tolerance, locality, circuit depth, measurement-based protocols
- Extend to non-CSS / subsystem / higher-dim / qudit codes & non-Abelian groups